

Positive Definiteness of K_d

Notation. Let $\mathbf{d} \in \mathbb{R}^n$ be the Lagrangian (finger-space) coordinates, $\mathbf{d}_{\text{ref}}$ the virtual equilibrium, $\delta \mathbf{d} = \mathbf{d}_{\text{ref}} - \mathbf{d}$ the deflection, $\mathbf{A} = ((\mathbf{P}\mathbf{J}_x)^*)^\top \eta \mathbf{J}_d^\top$, $\mathbf{v} = \mathbf{A}^\top \mathbf{n}$, $\mathbf{u} = \delta \mathbf{d}$, $\gamma = \|\mathbf{v}\|^2 \|\mathbf{u}\|^2$. We call K_d *positive definite* if its symmetric part $K_s := \frac{1}{2}(K_d + K_d^\top) \succ 0$.

1. Force-tracking update

The predicted force along contact normal $\mathbf{n}$ is $f = \mathbf{n}^\top \mathbf{A} K_d \delta \mathbf{d}$. The gradient of $\mathcal{L} = \frac{1}{2}(f - f_{\text{des}})^2$ with respect to K_d is $e \mathbf{v} \mathbf{u}^\top$, giving the update

$$K_d^{(t+1)} = K_d^{(t)} - \alpha e_t \mathbf{v} \mathbf{u}^\top, \quad e_t = f^{(t)} - f_{\text{des}}.$$

Theorem 1. Let $K_d^{(0)} \succ 0$ and $f_{\text{des}} > 0$. For any $\alpha \in (0, 2/\gamma)$, $K_d^{(t)} \succ 0$ for all $t \geq 0$ provided

$$\lambda_{\min}(K_s^{(0)}) \cdot \|\mathbf{v}\| \|\mathbf{u}\| > e_0. \quad (1)$$

Proof. Let $\mathbf{P} = \frac{1}{2}(\mathbf{v} \mathbf{u}^\top + \mathbf{u} \mathbf{v}^\top)$. At each step $K_s^{(t+1)} = K_s^{(t)} - \alpha e_t \mathbf{P}$, so by Weyl's inequality

$$\lambda_{\min}(K_s^{(t+1)}) \geq \lambda_{\min}(K_s^{(t)}) - \alpha |e_t| \lambda_+(\mathbf{P}) \geq \lambda_{\min}(K_s^{(t)}) - \alpha |e_t| \|\mathbf{v}\| \|\mathbf{u}\|,$$

since $\lambda_+(\mathbf{P}) = (\mathbf{v}^\top \mathbf{u} + \|\mathbf{v}\| \|\mathbf{u}\|)/2 \leq \|\mathbf{v}\| \|\mathbf{u}\|$. Telescoping and using $|e_t| = |e_0|(1 - \alpha\gamma)^t$:

$$\begin{aligned} \lambda_{\min}(K_s^{(T)}) &\geq \lambda_{\min}(K_s^{(0)}) - \alpha \|\mathbf{v}\| \|\mathbf{u}\| \sum_{t=0}^{T-1} |e_t| \geq \lambda_{\min}(K_s^{(0)}) - \underbrace{\alpha \|\mathbf{v}\| \|\mathbf{u}\| \cdot \frac{|e_0|}{\alpha\gamma}}_{= |e_0|/\|\mathbf{v}\| \|\mathbf{u}\|} \\ &= \lambda_{\min}(K_s^{(0)}) - |e_0|/(\|\mathbf{v}\| \|\mathbf{u}\|) \end{aligned}$$

The α cancels: the bound $\lambda_{\min}(K_s^{(0)}) - |e_0|/(\|\mathbf{v}\| \|\mathbf{u}\|)$ is independent of α and strictly positive under (1), uniformly in T . $\square$

Remark. Smaller α means smaller steps but more of them; the total variation $\alpha \sum_t |e_t| = |e_0|/\gamma$ is the same for all $\alpha \in (0, 2/\gamma)$. There is therefore no separate condition on α for positive definiteness.

Corollary 1. If $K_d^{(0)} = k_0 \mathbf{I}$, $k_0 > 0$, then $K_d^{(t)} \succ 0$ for all $t \geq 0$ and all $\alpha \in (0, 2/\gamma)$, unconditionally.

Proof. By Cauchy-Schwarz, $e_0 = k_0 \mathbf{v}^\top \mathbf{u} - f_{\text{des}} \leq k_0 \|\mathbf{v}\| \|\mathbf{u}\| - f_{\text{des}}$, so $e_0/(\|\mathbf{v}\| \|\mathbf{u}\|) \leq k_0 - f_{\text{des}}/(\|\mathbf{v}\| \|\mathbf{u}\|) < k_0 = \lambda_{\min}(K_s^{(0)})$, and condition (1) holds. $\square$

2. Feedforward stiffness inversion

The first-order tip stiffness is the congruence $K_x = \mathbf{M}^\top K_d \mathbf{M}$, $\mathbf{M} = \mathbf{J}_d(\mathbf{P}\mathbf{J}_x)^*$. Given $K_{x,\text{des}} \succ 0$ and invertible $\mathbf{M}$, the feedforward stiffness is $K_d = (\mathbf{M}^\top)^{-1} K_{x,\text{des}} \mathbf{M}^{-1}$.

Theorem 2. $K_d = (\mathbf{M}^\top)^{-1} K_{x,\text{des}} \mathbf{M}^{-1} \succ 0$ always.

Proof. For any $\mathbf{x} \neq \mathbf{0}$, set $\mathbf{y} = \mathbf{M}^{-1} \mathbf{x} \neq \mathbf{0}$. Then $\mathbf{x}^\top K_d \mathbf{x} = \mathbf{y}^\top K_{x,\text{des}} \mathbf{y} > 0$. $\square$